

Joe Worsfold

Founder & Principal Architect

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Founder and Technical Director at Stack Foundry UK. Principal Architect for agentic AI workflows and event-driven systems in Rust, TypeScript, and Python. AWS AI Award winner. I embed as a principal technical leader – or consult as an external architect – to ship autonomous, high-performance systems without sacrificing operational reliability.

Skills

Core

Rust TypeScript Python Go Agentic AI workflows RAG & embeddings AWS (serverless / event-driven)

Also strong

OpenSearch LangChain agents Neo4j DynamoDB / Postgres React / Next.js CDK / Terraform Cursor / AI-native engineering

Positions

Principal Architect Founding Engineer Staff AI

Experience level

Principal Lead Senior

Work eligibility

UK

Work type

Remote

Languages

English (fluent)

Experience

Founder & Principal Architect, Stack Foundry Ltd

07/2025 – Present
1 year 4 months

Remote

Founder-led consultancy. Principal architecture for agentic AI workflows and specialty-insurance data systems; building Prism and Helmsman.

Consultancy started while wrapping Aviva Zero.

Rust TypeScript Python AWS (Lambda OpenSearch event-driven) LangChain RAG/embeddings SurrealDB Cursor

- **Production MDM platform:** Delivered a production master data management platform for a specialty insurance client in six months, replacing multi-system complexity with one source of truth – fewer engineers to maintain and fewer ops agents to run. Used at production scale by underwriters and dataops.
- **RAG & semantic search:** Designed LLM selection, RAG, and embedding pipelines with OpenSearch semantic enrichment to improve discoverability across messy insurance master data.
- **Agentic entity resolution:** When deterministic entity resolution failed, introduced LangChain agentic workflows so the system could gather context and propose decisions for human approval – increasing agency without silent autonomous binds.
- **AI-native engineering:** Taught client developers AI-native delivery and built bespoke guardrails for agentic coding workflows (Cursor-centred practice).
- **Prism workbench:** Building Prism – an agent-to-agent workbench focused on local context assembly and document embeddings for underwriting-style knowledge work (active product development; not a customer production deploy).
- **Helmsman:** Created Helmsman – rules, skills, and checks that score AI-readiness and keep agent-written code honest. Used on Prism and NIDOS; patterns inspired by production AI delivery flows.

Senior Full Stack Software Engineer, Aviva Zero

01/2024 – 11/2025
1 year 10 months

London

Climate-focused insurance platform — high-availability Go/TypeScript services and accessible React frontends.

Backend: Go, AWS Lambda, API Gateway, EventBridge, SQS

Frontend: TypeScript, React, Next.js, ECS | CI/CD: CDK, GitHub Actions, Datadog, PagerDuty

- **Event-driven serverless:** Designed and implemented event-driven serverless architectures in Go using Lambda, SQS, EventBridge, and SES for asynchronous workflows, scheduling, and alerting across policy and payment services.
- **Scale & availability:** Built and maintained a platform that has sold over a million policies and supports tens of thousands of daily users, with 24/7 reliability through monitoring and performance work.
- **Financial services:** Delivered financially sensitive services including BACS payment workflows, interest-bearing monthly instalments, and secure customer interactions within FCA-regulated boundaries.
- **Observability:** Led alerting and observability improvements for proactive detection and resolution.
- **Accessible UI:** Shipped accessible React frontends from Figma using Chakra UI and Storybook, optimised via multivariate testing.
- **Advisory:** Advised the transition from start-up to enterprise-scale systems and helped structure product squad roadmaps.

Lead Software Engineer (Tech Lead), Flock Cover

02/2023 – 01/2024

11 months

London

Next-generation insurance platform with modern cloud and serverless infrastructure.

Backend: TypeScript, Scala, Lambda, API Gateway, EventBridge, SQS

Frontend: React, Next.js, ECS/Lambda | IaC: CDK, Terraform, GitHub Actions, Datadog

- **Insurance platform:** Led full-stack development across Node/TypeScript backends and React/Next.js frontends with a focus on scalability and performance.
- **Microservices:** Designed microservices on Lambda, ECS, and EventBridge for observable, 24×7 services that scale with demand.
- **Infrastructure as Code:** Championed Terraform and CDK for repeatable AWS provisioning.
- **Team leadership:** Mentored engineers and raised the bar on code quality and infrastructure practice.
- **CI/CD:** Streamlined CI/CD with GitHub Actions for reliable production deploys.

Senior Consultant Full Stack Engineer, Infinity Works (part of Accenture)

10/2021 – 02/2023

1 year 4 months

UK

Tech lead for specialty-insurance automation and serverless integrations (client work anonymised where required).

TypeScript | Node.js | Postgres | DynamoDB | Lambda | API Gateway | Step Functions | EventBridge | Terraform | Datadog

- **Underwriting automation:** Led automation for underwriting data extraction and distribution with TypeScript, Node.js, and Postgres on AWS.

- **Event-driven workflows:** Built serverless, event-driven integrations with Step Functions and EventBridge for multi-service workflows.
- **Tech leadership:** Acted as Tech Lead – unblocked hard engineering problems, ran knowledge-sharing, and strengthened engineering culture.
- **Serverless microservices:** Architected Lambda / API Gateway / DynamoDB microservices with CloudWatch and Datadog observability.

Software Engineer, UK Home Office

09/2020 - 10/2021
1 year 1 month

UK

Graph-backed intelligence tooling and government web applications.

React · Angular · Node.js · MongoDB · Neo4j · Docker · Kubernetes · Drone

- **Graph intelligence:** Supported ETL into Neo4j graph databases modelling people, assets, and movements; wrote performant Cypher for pattern discovery at scale – central to the platform's intelligence capabilities.
- **Operationalising ML:** Worked with data scientists to operationalise ML against the graph layer for anomaly detection and higher-quality investigative leads.
- **Government applications:** Built React/Angular apps and Node REST APIs with TDD/BDD, Docker, and Kubernetes CI via Drone.

IT Developer, Mott MacDonald

09/2018 - 09/2020
2 years

UK

ERP integration and process automation.

Java · JavaScript · PL/SQL

- **ERP integration:** Built Java integrations from legacy systems to cloud ERPs using SOLID, design patterns, and TDD.
- **Process automation:** Delivered electronic forms/workflows and PL/SQL-backed data solutions.

Selected projects

Tooling Helmsman

Open rules, skills, and checks for agent-written code – readiness scoring, review, and STAR-stack guardrails. Used on Prism and NIDOS.

Context

Hack Hate 2020: Hate Detector

Winner – Best use of AI/ML (AWS) and Best use of location (ESRI). Processed and classified 18.5M+ tweets; graph model in Neo4j for hate-speech network visualisation.

Presentation · Code

Education

Birkbeck, University of London

2017 - 2019

MSc Computer Science, Distinction

Algorithms, data structures, machine learning, and NLP with emphasis on scalable AI-driven systems.

Thesis: Character-level Convolutional Neural Networks for Hate Speech Detection with Intelligent Adversaries – CNN text classification at scale with graph modelling of classified content in Neo4j.

Royal Holloway, University of London

2009 – 2013

BA History, 2:1

Analytical writing, research, and communication – foundations that still show up in product and stakeholder work.